

Effective simplex rigidity for the projection cone invariant

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Abstract

For every fixed dimension $d \geq 2$, we give a fully explicit power modulus for the lower equality case of the projection-cone invariant. If $e(K) = a(K) - 1/(d+1)$ and S is any maximum-volume simplex inscribed in K , with centroid z , then

$$S \subseteq K \subseteq z + (1 + G_d e(K)^{\theta_d})(S - z), \quad \theta_d = \frac{1}{d(d^2 + 2d + 2)}.$$

All constants are specified. The proof works for arbitrary convex bodies and cone-volume laws, including nonatomic laws. Its constants are extremely poor and no optimality or novelty is asserted. In dimension two the exact identity $a(K) = 2|K|/|K - K|$ connects the question to classical Rogers–Shephard stability: an estimate in Böröczky’s work already implies the much stronger linear bound $d_{BM}(K, T_2) \leq 1 + 576e(K)$. Only the simplex endpoint is treated here; version 4 concerns the symmetric product endpoint.

1. Statement and scope

A convex body is a compact convex set with nonempty interior. For $K \subset \mathbb{R}^d$, let $|K|$ denote volume and let ΠK be its projection body, normalized by

$$h_{\Pi K}(u) = |K| |u^\perp|_{d-1} \quad (|u| = 1).$$

Put

$$R(K) = \frac{|\Pi K|}{|K|^{d-1}}, \quad a(K) = \left(\frac{d}{d+1} \right)^d \frac{R(\mathcal{P}K)}{R(K)} - 1,$$

where $\mathcal{P}K$ is a pyramid over K . Versions 2–3 establish that a is a continuous affine invariant, give its cone-volume representation recalled below, and prove $a(K) \geq 1/(d+1)$ with equality exactly for simplices. We use those identities and standard Cauchy and mixed-volume formulas. The full quantitative argument is supplied here.

For any maximum-volume inscribed simplex S with centroid z , define

$$E(K, S) = \inf\{t \geq 0 : K \subseteq z + (1+t)(S - z)\}.$$

The multiplicative affine Banach–Mazur distance from K to a simplex is the infimum of dilation factors for a sandwich between an inscribed simplex and a translate of its dilate. In particular it is at most $1 + E(K, S)$.

Explicit constants. For $d \geq 2$, define, in this order,

$$\begin{aligned} R_0 &= d(d+1), & b &= \frac{1}{4(dR_0)^d}, & M &= \frac{2}{b}, \\ L &= d2^d, & Z &= M(2d+3), & C &= 4d^2R_0(M+1), \\ p &= d^2 + 2d + 2, & \theta_d &= \frac{1}{dp}, & \chi &= b^d \left(\frac{b^d}{12dL} \right)^{d(d+1)}. \end{aligned}$$

Next set

$$\begin{aligned} \eta_0 &= \min \left\{ \frac{1}{2(d+1)}, \frac{b}{4d}, \frac{1}{Z}, \frac{1}{Z(8MdC)^d} \right\}, \\ e_0 &= \frac{\chi \eta_0^p}{2(d+1)}, \\ A_d &= 4MdCZ^{1/d} \left(\frac{2(d+1)}{\chi} \right)^{\theta_d}, \\ G_d &= \max\{A_d, (R_0 - 1)e_0^{-\theta_d}\}. \end{aligned}$$

These are explicit numbers depending only on d ; χ, η_0, e_0 are rational.

Theorem 1 (effective simplex rigidity). For every convex body $K \subset \mathbb{R}^d$, $d \geq 2$, and every maximum-volume inscribed simplex S ,

$$E(K, S) \leq G_d e(K)^{\theta_d}, \quad e(K) = a(K) - \frac{1}{d+1}.$$

If $0 \leq e(K) \leq e_0$, the sharper stated constant applies:

$$E(K, S) \leq A_d e(K)^{\theta_d}, \quad A_d e_0^{\theta_d} \leq \frac{1}{2}.$$

Consequently

$$\delta(d, \varepsilon) = \min\{e_0, (\varepsilon/A_d)^{dp}\}$$

is an explicit choice in version 3's qualitative containment theorem. In dimension one K is itself an interval and $E(K, S) = 0$.

The first exponents are $1/20, 1/51, 1/104$ in dimensions $2, 3, 4$. Already for $d = 2$ the displayed local threshold satisfies $10^{-279} \leq e_0 < 10^{-278}$. The theorem supplies an effective modulus, not a practical numerical estimate. Section 8 explains the stronger inherited planar result. No dimension-independent conclusion follows.

2. Normalization and the exact cancellation defect

Fix an arbitrary maximum simplex S . Map it affinely to a regular simplex Δ with centroid zero and inradius one. Its vertices $v_0, \dots, v_d$ have norm d . If $\alpha_i(x)$ are barycentric coordinates relative to Δ , replacing v_i by $x \in K$ multiplies volume by $|\alpha_i(x)|$. Maximality gives $|\alpha_i(x)| \leq 1$, hence

$$B_2^d \subseteq \Delta \subseteq K \subseteq R_0 B_2^d. \tag{2.1}$$

The deficit is unchanged by this normalization.

Write $h = h_K$ and let S_K denote surface area measure. Push the probability measure

$$\frac{h(u)}{d|K|} dS_K(u)$$

forward by $u \mapsto u/h(u)$, obtaining ν . This centered law is supported on $\partial K^\circ \subseteq B_2^d$. For independent samples X_i of law ν , put

$$A = \mathbb{E}|\det(X_1, \dots, X_d)|, \quad B = \mathbb{E}|\det((X_1, 1), \dots, (X_{d+1}, 1))|.$$

The cone-volume representation from version 3 reads

$$a(K) = \frac{B}{(d+1)A}, \quad D := B - A = (d+1)Ae(K) \leq (d+1)e(K), \quad (2.2)$$

because $0 < A \leq 1$. For every unit vector u , centering and Cauchy's projection formula give

$$\int (-\langle u, x \rangle)_+ d\nu(x) = \frac{|K| |u^\perp|_{d-1}}{d|K|} \geq \frac{(2/d)^{d-1}}{d(2R_0)^d} = 2b. \quad (2.3)$$

Here the projection contains the unit ball in dimension $d-1$, which contains a cube of side $2/d$.

For an ordered base tuple $\mathbf{x} = (x_1, \dots, x_d)$ define

$$F_{\mathbf{x}}(y) = \det((x_1, 1), \dots, (x_d, 1), (y, 1)) = \det(x_1 - y, \dots, x_d - y),$$

and $P(\mathbf{x}) = \int (F_{\mathbf{x}})_+ d\nu$, $N(\mathbf{x}) = \int (-F_{\mathbf{x}})_+ d\nu$. Centering gives

$$P(\mathbf{x}) - N(\mathbf{x}) = \det(x_1, \dots, x_d).$$

Thus the exact cancellation identity is

$$D = 2 \int \min\{P(\mathbf{x}), N(\mathbf{x})\} d\nu^{\otimes d}(\mathbf{x}). \quad (2.4)$$

This identity includes horizontally singular base tuples.

If two $(d+1)$ -tuples in B_2^d have corresponding points at distance at most r , their affine determinants differ by at most

$$Lr = d2^d r. \quad (2.5)$$

Indeed, write it using the d difference columns above and telescope one column at a time. Each old or new column has norm at most two and each change has norm at most $2r$.

3. Concentration on heavy cells

Fix $0 < \eta \leq \eta_0$ and set

$$r = \frac{\eta b^d}{4L}, \quad \tau = \eta \left(\frac{r}{3d} \right)^d, \quad \gamma = \eta b^d. \quad (3.1)$$

Partition $[-1, 1]^d$ into measurable cubes of diameter at most r , assigning boundaries to single cells. Dividing each axis into $\lceil 2\sqrt{d}/r \rceil$ intervals uses at most $(3d/r)^d$ cells, since $r \leq 1$. Call a cell heavy if its ν -mass is at least τ . The light cells together have mass at most η .

Choose one support point in each heavy cell and let X be this finite set. The chosen points need not have positive individual mass. The function $(-\langle u, x \rangle)_+$ is one-Lipschitz and bounded by one on the support. Discarding light cells and replacing samples by representatives therefore gives

$$\sum_{Q \text{ heavy}} \nu(Q)(-\langle u, x_Q \rangle)_+ \geq 2b - \eta - r \geq b.$$

The weights sum to at most one, so for every unit u some representative has $\langle u, x_Q \rangle \leq -b$. Hence

$$bB_2^d \subseteq \text{conv } X. \quad (3.2)$$

Choose a maximum-volume simplex $T = \text{conv}(w_0, \dots, w_d)$ with vertices in X , and put

$$V = |\det((w_0, 1), \dots, (w_d, 1))|.$$

Then $V \geq b^d$: the simplex with vertices $0, be_1, \dots, be_d$ lies in $\text{conv } X$, and separate convexity of the absolute affine determinant lets each of its vertices be replaced successively by a point of X without decreasing the determinant. At this stage we do not assume that $0 \in T$.

Write $x = \sum_i \alpha_i w_i$ for $x \in X$, with $\sum_i \alpha_i = 1$. There are two possible robust witnesses:

1. If some $\alpha_i \leq -\eta$, use the d vertices other than w_i as base. The determinant values at w_i and x have opposite signs and magnitudes at least ηV .
2. If $\alpha_i, \alpha_j \geq \eta$ for two distinct indices, use x and the $d-1$ original vertices other than w_i, w_j as base. Expansion of $(x, 1)$ gives opposite determinant values at w_i, w_j , of magnitudes $\alpha_j V$ and $\alpha_i V$.

In each case the d base representatives and the two tested representatives lie in $d+2$ distinct heavy cells. Since $Lr = \gamma/4$ and $\eta V \geq \gamma$, inequality (2.5) preserves both signs with margin at least $\gamma/2$ for every tuple of actual support points in the corresponding cells. For each sampled base tuple, both integrals in (2.4) are at least $\tau\gamma/2$. The ordered product of base cells has mass at least τ^d . Consequently either witness forces

$$D \geq \tau^{d+1}\gamma = \chi\eta^p. \quad (3.3)$$

This integrates over full positive-measure cells, not over the selected points.

Assume now that

$$D < \chi\eta^p. \quad (3.4)$$

Both witnesses are impossible. Every representative has all coefficients greater than $-\eta$ and at most one coefficient at least η . There is exactly one such coefficient since $\eta \leq 1/[2(d+1)]$. The other d coefficients have absolute value at most η , whence

$$\text{dist}(x, \{w_0, \dots, w_d\}) \leq 2d\eta \quad (x \in X).$$

Combining this with (3.2), support functions and $2d\eta \leq b/2$ proves

$$\frac{b}{2}B_2^d \subseteq T \subseteq B_2^d. \quad (3.5)$$

Thus zero is now known to be interior to T . Assign a sample in each heavy cell to its nearby vertex and assign light-cell samples arbitrarily. The resulting expected distance satisfies

$$\int \text{dist}(x, \{w_0, \dots, w_d\}) d\nu(x) \leq r + 2d\eta + 2\eta \leq (2d + 3)\eta. \quad (3.6)$$

4. An enclosing simplex and a volume estimate

Let

$$P = T^\circ = \{y : \langle w_i, y \rangle \leq 1 \text{ for all } i\}.$$

Each w_i belongs to ∂K° . By (3.5), P is a bounded simplex and

$$K \subseteq P \subseteq MB_2^d. \quad (4.1)$$

Every w_i is a vertex of T , so $h_P(w_i) = 1$. Also $h_K(x) = 1$ on $\text{supp } \nu$. The support function h_P is M -Lipschitz, and (3.6) implies

$$0 \leq \int (h_P(x) - 1) d\nu(x) \leq Z\eta. \quad (4.2)$$

The left-hand side equals

$$\frac{1}{d|K|} \int h_P(u) dS_K(u) - 1 = \frac{V(K[d-1], P)}{|K|} - 1.$$

Minkowski's first inequality therefore gives

$$|P|^{1/d} \leq |K|^{1/d}(1 + Z\eta).$$

Since $Z\eta \leq 1$,

$$|P| - |K| \leq (2R_0)^d d2^{d-1} Z\eta. \quad (4.3)$$

The conversion uses the constructed circumscribed simplex directly; it does not require an inverse-continuity constant for surface area measures.

5. A cap estimate

Put $s = d_H(K, P)$. If $s > 0$, choose a farthest point $q \in P$ from K and the unit direction u from its metric projection onto K , so

$$\langle u, q \rangle - h_K(u) = s.$$

Set $t = s/[2(M+1)]$. We have $s \leq M$, hence $0 < t < 1$. The ball

$$(1-t)q + tB_2^d$$

lies in $\text{conv}(q, B_2^d) \subseteq P$. Its lowest u -coordinate is at least $h_K(u) + s/2$, because $\langle u, q \rangle \leq M$. It is therefore disjoint from K . Its inscribed cube has side $2t/d$, giving

$$|P| - |K| \geq \left(\frac{s}{d(M+1)} \right)^d.$$

Together with (4.3), this yields, also when $s = 0$,

$$s \leq 2dR_0(M+1)[d2^{d-1}]^{1/d}(Z\eta)^{1/d} \leq C(Z\eta)^{1/d}. \quad (5.1)$$

The last term in the definition of η_0 ensures

$$s \leq \frac{1}{8Md}. \quad (5.2)$$

6. Returning to the original maximum simplex

The conclusion must concern the original arbitrarily chosen Δ , not just the simplex P constructed from the law. Since $h_P \geq 1$ on the unit sphere and $h_K \geq h_P - s$,

$$(1-s)P \subseteq K \subseteq P.$$

Maximality of Δ then gives

$$\frac{|\Delta|}{|P|} \geq (1-s)^d \geq 1-ds.$$

Put $\delta = ds \leq 1/8$. The barycentric coordinates of the $d+1$ vertices of Δ relative to P form a nonnegative column-stochastic matrix W satisfying

$$|\det W| = \frac{|\Delta|}{|P|} \geq 1-\delta. \quad (6.1)$$

Each column has Euclidean norm at most one. Hadamard's inequality forces the squared norm of each column to be at least $(1-\delta)^2$, so its largest entry is at least $1-2\delta$.

The dominant rows are distinct. Indeed, the permanent is the probability that independent categorical draws from the columns choose distinct rows. A common dominant row in two columns would imply

$$|\det W| \leq \text{per } W \leq 1 - (1-2\delta)^2 \leq 4\delta < 1-\delta,$$

contrary to (6.1). Relabel the vertices of P accordingly. Each vertex of Δ is within

$$2\delta \text{diam } P \leq 4M\delta$$

of its matched vertex of P . Taking convex combinations proves

$$d_H(\Delta, P) \leq 4Mds.$$

Because $B_2^d \subseteq \Delta$ and Δ has centroid zero,

$$K \subseteq P \subseteq \Delta + 4MdsB_2^d \subseteq (1+4Mds)\Delta.$$

We have proved the parameterized estimate

$$E(K, \Delta) \leq 4MdC(Z\eta)^{1/d} \quad (6.2)$$

whenever $0 < \eta \leq \eta_0$ and $D < \chi\eta^p$.

7. Completing the explicit modulus

If $0 < e = e(K) \leq e_0$, choose

$$\eta = \left(\frac{2(d+1)e}{\chi} \right)^{1/p}.$$

Then $\eta \leq \eta_0$ and (2.2) gives

$$D \leq (d+1)e = \frac{\chi}{2}\eta^p < \chi\eta^p.$$

Substituting in (6.2) proves $E(K, \Delta) \leq A_d e^{\theta_d}$. At $e = 0$, the parameterized bound holds for every sufficiently small positive η ; letting $\eta \downarrow 0$ proves the same assertion. No equality conclusion is obtained by merely passing to a limit of strict inequalities.

At the threshold,

$$A_d e_0^{\theta_d} = 4M d C (Z\eta_0)^{1/d} \leq \frac{1}{2}.$$

For $e > e_0$, (2.1) gives $K \subseteq R_0 \Delta$, hence

$$E(K, \Delta) \leq R_0 - 1 \leq (R_0 - 1)e_0^{-\theta_d} e^{\theta_d}.$$

This proves the global constant G_d . Undo the initial affine map: the centroid goes back to z , and the homotheties become those about z . The original maximum simplex was arbitrary and all constants depend only on d , so the result holds for every such simplex. This completes the proof of Theorem 1.

8. The planar identity and stronger inherited stability

Proposition 2 (planar reduction). Every planar convex body satisfies

$$a(K)R(K) = 2, \quad R(K) = \frac{|K - K|}{|K|}.$$

Proof. First take a polygon with the origin in its interior. List its edges counterclockwise. Let u_i be the edge length times outward unit normal, b_i the edge length times support number, and $w_i = (u_i, b_i)$. Then $b_i > 0$ and u_i/b_i are the counterclockwise polar vertices. Hence for $i < j < k$ all lifted determinants have positive sign, and the numerator in the polytope formula for a is

$$\Sigma = \sum_{i < j < k} \det(w_i, w_j, w_k).$$

Let $T_0 = \sum_{i < j} \det(u_i, u_j)$. On expanding Σ , the coefficient of b_k is

$$\sum_{i < j < k} \det(u_i, u_j) - \sum_{i < k < j} \det(u_i, u_j) + \sum_{k < i < j} \det(u_i, u_j).$$

It equals T_0 . To verify this, put $U = \sum_{i < k} u_i$ and $W = \sum_{j > k} u_j$. Closure gives $W = -U - u_k$, so

$$\det(U, W) = -\det(U, u_k), \quad \det(u_k, W) = \det(U, u_k),$$

which cancels the difference between the two expressions. The vectors u_i are the oriented edges rotated by a fixed quarter-turn; the signed area formula gives $T_0 = 2|K|$. Also $\sum_i b_i = 2|K|$. Thus $\Sigma = 4|K|^2$.

The denominator in the polytope formula from version 2 is $2|K||\Pi K|$, so $a(K) = 2|K|/|\Pi K|$. In the plane the support function of ΠK is the width in a perpendicular direction; consequently ΠK is a quarter-turn of $K - K$. This proves both identities for polygons. Polygonal approximation and continuity extend them to every planar convex body.

Put $e = a(K) - 1/3$ and $\rho = 1 - R(K)/6$. Proposition 2 gives

$$\rho = \frac{3e}{1 + 3e}.$$

Theorem 1 of Böröczky [B] already gives a linear Banach–Mazur stability estimate. More sharply, Section 8 of the inspected author-hosted text gives $R(K) \leq 6 - t/32$ when $d_{BM}(K, T_2) = 1 + t$. Hence the inherited consequence is

$$d_{BM}(K, T_2) - 1 \leq 192\rho = \frac{576e}{1 + 3e} \leq 576e.$$

This prior planar result is stronger in exponent than Theorem 1’s planar Banach–Mazur consequence. We do not present planar stability as a new theorem. The reduction to a difference body is specific to dimension two; no such formula is asserted in higher dimensions.

9. Relation to version 4 and limitations

Version 4 classifies the centrally symmetric upper endpoint: $a(K) = 1/2$ holds exactly, up to an invertible linear map, for Cartesian products of centrally symmetric bodies of dimensions one and two. It also gives a fixed-dimensional qualitative stability statement for that class. The present theorem treats the lower deficit $a(K) - 1/(d + 1)$, whose equality class consists of simplices. It does not produce a quantitative modulus for $1/2 - a(K)$, and it does not determine the spectral supremum discussed in versions 2–3.

No boundary smoothness or atom assumption occurs in the proof. The quantitative witness uses $d + 2$ positive-mass cells; all determinant signs are made uniform on those cells before integration. The losses in the exponent come from the covering, the $d + 1$ cell-mass factors, and the cap estimate. No sharpness or favorable dependence on dimension is claimed. The previously documented family $T_m \times T_m$ remains a barrier to a dimension-independent additive-defect modulus.

Quantitative random-simplex stability [AB], cone-volume stability [BH], and inverse stability for surface area measures [HS] are established subjects. The particular cone-law deficit and elementary enclosing-simplex argument are the formulation proved here, not evidence that it lacks antecedents. Saroglou’s cone/projection-body work [S] is especially relevant; only its publisher abstract and author listing were available in this comparison, so formula-by-formula overlap remains unchecked. This is a bounded literature comparison, not a priority or best-known-rate claim.

The standard-library exact checks accompany the written proof. They verify the displayed constants in dimensions 2 through 8, 400 determinant perturbations, 400 barycentric matrices, 8,320 nonatomic box-corner sign tests, and 200 polygon identities. Finite checks are regression evidence, not a proof of the general theorem. Independent model audits are not external human peer review or proof-assistant formalization.

References and provenance

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Versions 2–4 preserve and credit the product and simplex computations inherited from OpenAI family 088, pinned at commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. This supplement adds an explicit quantitative proof to that continuing record; it does not replace historical versions. The manuscript is AI-assisted research and makes no first-discovery assertion.